

Pooja Kiran

Security Engineer | Agent Security | Application Security | Cloud IAM

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EXPERIENCE

Independent AI Security Researcher & Engineer

Aug. 2024 - Present

Self-Directed Research

Tempe, AZ

- Built the OS Resource Management Simulator in Flask and Docker, then evolved it into a PostgreSQL-backed lease service with atomic acquisition, expiring leases, idempotent retries, role-based API access, audit history, metrics, and concurrency validation.
- Expanded into cloud and identity security by applying AWS IAM, least-privilege access, policy evaluation, and CloudTrail-backed auditability, building the foundation for later identity and authorization tooling.
- Extended the work into AI and agent security, engineering controls for runtime tool calls, IAM analysis, model provenance, and training-data integrity while standardizing evidence through CI, SARIF/Code Scanning, Elastic Security, Prometheus, Docker, and reproducible tests.

Business & Compliance Lead, AEROSEC

Aug. 2025 - Dec. 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ

- Led business/compliance analysis for a proposed airline PSS protection layer and built a **\$120K first-year commercialization scenario** with a five-year financial model presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant, IT Grader

Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments**, giving evidence-based feedback on security controls, policy, compliance, risk analysis, and implementation quality.

PROJECTS

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security*

Jul. 2026 - Present

- Secured the agent-to-tool execution boundary by engineering a default-deny MCP/JSON-RPC gateway with capability authorization, PII/exfiltration checks, anti-SSRF controls, rate limiting, and fail-closed enforcement before downstream execution.
- Expanded runtime detection with **55 prompt-injection patterns** and tamper-evident audit logging, producing reviewable security telemetry for policy decisions and incident analysis.
- Validated the gateway with **718 passing tests at 82.46% coverage**, including denied-call enforcement, and implemented **9 Elastic rules with 21 SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0*

Aug. 2026 - Sep. 2026

- Reduced over-privileged agent and workload identity risk by building a static IAM analyzer with **25 deterministic rules** covering wildcard grants, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, trust-policy weaknesses, audit tampering, and permission boundaries.
- Made IAM findings reviewable in CI by emitting text, JSON, and SARIF 2.1.0 results into GitHub Code Scanning, turning policy weaknesses into actionable code-review evidence.
- Verified rule behavior and failure paths with **235 passing tests**, while keeping credential-dependent live-scan checks isolated from deterministic static-analysis coverage.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras*

Jul. 2026 - Sep. 2026

- Reduced model supply-chain exposure by developing a non-executing scanner that inspects untrusted repositories and artifacts without importing or running their code.
- Combined pickle-opcode, AST/taint, dependency/provenance, obfuscation, and binary-format checks to surface risky behavior across SafeTensors, GGUF, ONNX, Keras, and pickle-derived artifacts.
- Verified **241 passing tests at 75.81% coverage**, detected **33/33 committed adversarial fixtures**, and recorded **0 actionable findings across 4 committed benign samples**.

EDUCATION

Arizona State University

Tempe, AZ

M.S., Information Technology (Security), GPA: 3.87/4.00

Aug. 2024 - May 2026

M. S. Ramaiah University of Applied Sciences

Bengaluru, India

B.Tech., Computer Science & Engineering, CGPA: 8.44/10

Aug. 2019 - Aug. 2023

TECHNICAL SKILLS

Agent & AI Security: MCP/JSON-RPC 2.0, Tool/Function-Call Security, Prompt Injection, LLM Red Teaming, Model Supply-Chain Security, Training-Data Integrity, Adversarial ML

Cloud & Application Security: AWS IAM, Least Privilege, Trust Policies, Permission Boundaries, Threat Modeling, API Security, Capability-Based Authorization, Data Exfiltration Detection

Engineering & Detection: Python, Rust, C++, FastAPI, pytest, Hypothesis, GitHub Actions, SARIF 2.1.0, CodeQL, Docker, Kubernetes, Elastic Security, Prometheus

CERTIFICATIONS

AWS Academy Graduate - Cloud Architecting (Apr. 2025) | AWS Academy Graduate - Cloud Security Foundations (Nov. 2025)